Thesis Outline

Topic: “Quantum Error Correction with Rydberg Atom based Quantum Systems”

1. Abstract
2. Theoretical Foundations
    a. Rydberg Physics
        i. Introduction
        ii. Interacting Rydberg atoms : Dipole-Dipole and Van der waals interactions
        iii. Rydberg Blockade and Rydberg Faciliation Mechanism
    b. Light Matter Interaction : Open Quantum Systems
        i. Introduction
        ii. Lindbald Master Equation
        iii. Rapid Adiabatic Passage (RAP) / Landau Zener Transition
    c. Quantum Error Correction
        i. Introduction
        ii. Repetition Codes - Bit Flip Correction Code
        iii. Repetition Codes - Phase Flip Correction Code
        iv. Shor’s Code
        v. Topological Surface Code
3. Benchmarking Physical Processes
    a. Open system dynamics of toy 2 - atom Rydberg System
        i. System Description (Hamiltonian, Parameters, ...)
        ii. Effect of Decay, Dephase
    b. Rapid Adiabatic Passage (RAP)
        i. 2 - atom system
        ii. 3 - atom system
            1. Site Dependent Detuning
            2. Site Independent Detuning
    c. RAP for different detuning rates
        i. In presence of decay
        ii. In presence of dephase
    d. Monte Carlo Wave Function Method (MCWF)
4. Bit Flip Code - 3+2 atom Rydberg system
    a. Parameter Optimization (All parameters including decay conditions)
    b. Qubit States, System Architecture
    c. Protocol Workflow and Hamiltonian Design
    d. Results
5. Phase Flip Code - 3+2 atom Rydberg system
    a. Encoding and Dynamical Phase Shift - The Problem
    b. Redefining qubit states, System Architecture
    c. Protocol Workflow and Hamiltonian Design
    d. Results
6. Shor’s Code (yet to be started)
7. Topological Surface Code (yet to be started..hopefully)
8. Conclusion